Research Note: The Physics of Coexistence — Force, Charge, Mass, and Time in Satta–Unit Interaction

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Status: Internal research note (not a catalog entry). Compiled to support drafting on *The Ontology of Coexistence* §1.2 and related studies.

Scope: Every explicit statement located in the primary texts (SB, MVD, JV, AVD, JVD, KD) that uses physics vocabulary — pressure, motion, force, magnetism, electricity, heat, charge/excitation (*āvesh* — context-dependent, see §8), mass, weight, gravitation, atomic structure (nucleus, orbits, elements), states of matter, density, radioactivity, indestructibility/conservation, cosmogony, time — to describe either (a) how *satta* (Omnipotence/Omnipresence) and units (*ikai*) relate, or (b) how units relate to one another. Citations use the project's tag system (SB \= *Samadhanatmak Bhautikvad* / "Resolution Centred Materialism"; MVD \= *Madhyasth Darshan Coexistentialism*; JV \= *Jeevan Vidya: An Introduction*; AVD \= *Adhyatmvad*; JVD \= *Janvad*; KD \= *Manav Karm Darshan*, Hindi-only — KD quotes are our working translations; full chapter translations now at `References/Madhyasth-Darshan/KD-English/`, see Method note) with page numbers verified directly against the source PDFs (see Method note at the end).

KD's stated core principle (*siddhant*, title page) is **śram–gati–pariṇām** — effort–motion–result — the same triple SB p. 62 unfolds from saturation (§4). Its section 3 (chapters 3.1–3.15) is the most physics-dense stretch anywhere in the corpus: atomic structure (ch. 3.1, p. 53), development in the atom (3.2, p. 59), heat and the Earth's balance (3.7, p. 72), state–motion (3.8, p. 83), quantity (3.9, p. 87), force–power (3.11, p. 101), reflection–projection (3.12, p. 109), pressure/flow/wave/electromagnetic force (3.13, p. 117), place–direction–distance–area–angle (3.14, p. 124), and time (3.15, p. 133).

This is a compilation of what the texts say, not yet a polished comparative argument. Read it as raw material — quotes first, brief glosses second — organized so the underlying picture becomes visible on its own.

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1. The base claim: two poles, one coexistence

Everything below sits on top of one recurring assertion: existence has exactly two aspects — formless, actionless *satta* and formful, active *prakriti* (nature, as countless units) — and neither produces the other.

"What is evident is that consciousness and matter are inseparably present. Upon examining their fundamental nature, we learn that all of existence is essentially nature (matter) saturated in Omnipotence (consciousness).” — SB, p. 48

“Omnipotence is formless and omnipresent. Within Omnipotence, nature is endowed with form, properties, essential nature, and dharma, and is inseparable from it. There is no place where Omnipotence is absent… Nature, saturated in Omnipotence, exists as countless units. Each unit, being saturated in Omnipotence, remains surrounded, submerged, and soaked in it.” — SB, p. 48

The relation-word the texts use for this bond is *samprikt* — saturated: soaked, submerged, surrounded. It is not “created by,” “emitted from,” or “caused by.” Whatever physical-sounding vocabulary follows (force, energy, charge, weight) has to be read against this base claim, because the texts are explicit that *satta* itself never acts.

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2. Satta is explicitly denied motion, pressure, and change

This is the texts’ own starting point for physical vocabulary — not an inference the study is making, but a direct statement, repeated in at least three of the five sources.

“Omnipotence remains in a state free from motion and pressure, whereas the whole of nature existing in Omnipotence is present with motion and pressure. Pressure refers to the compulsion towards attraction and repulsion arising under environmental influence (mutuality).” — SB, p. 49

“Consciousness, or knowledge, is omnipresent. It is permeative, transparent, and non-activity (devoid of wave or pressure). Since consciousness is omnipresent, it doesn’t prove to be a unit. Consciousness doesn’t prove to be the root cause of the universe’s origin, because it is devoid of change. Coexistence is the root cause of the universe. All the changes are only within nature saturated in consciousness.” — MVD, p. 74

“Formless existence in itself is the grandeur of absolute energy. It is in an eternally present state, without motion, wave, or pressure.” — SB, p. 50

Reading: pressure, attraction, and repulsion — the vocabulary of physical force — are placed explicitly at the level of unit-to-unit mutuality (“arising under environmental influence”), never as something *satta* does to a unit. This is the texts’ own firewall between the ground and physical action.

2.1 The same firewall applied to cosmogony: the Big Bang is rejected by name. JV contains the corpus’s one direct engagement with a named theory of modern cosmology, and it deploys exactly the §2 doctrine against it:

“How does nature get inspired by Omnipotence? Does it apply a force? **There is a hypothesis that it all started with a Big Bang, and the resulting force initiated everything one by one. Omnipotence (Space) does not possess any attribute to apply force.** There are no waves, motion, or pressure in Space, so the idea of applying force doesn’t arise. Every entity is energized, active, and inspired within Omnipotence, and this is evident… Units are distinct from one another because Omnipotence exists between them. If Omnipotence were not present, they wouldn’t be distinct. The joining of units occurs because

Omnipotence exists in between. This way, there is a basis for the composition and decomposition of entities in nature." — JV, p. 149

Reading: the Big Bang is not disputed on observational grounds (redshift, background radiation, etc. go unmentioned) but ruled out *a priori*: any cosmogonic first-force would have to be a force applied by the ground itself, and the ground is definitionally forceless. Note this sits on the same pages as the zero-attraction passage (§10.1) — in the source they are one continuous argument. The passage also restates the "distance is satta" claim of SB p. 79 (§4) from the other direction: satta-in-between is simultaneously what keeps units distinct *and* what lets them join. This pairs with §11 below (no creation, no annihilation): the darshan's alternative to cosmogony is not a different origin story but the denial that an origin event is needed at all — coexistence is *anaadi*, beginningless.

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3. Two kinds of energy: Absolute vs. Relative

MVD names this distinction directly, and it is probably the single most useful primitive for organising everything else in this note.

"Relative Energy (*sapeksh oorja*): The power which doesn't become apparent without the mutuality of units is referred to as relative energy. The powers, such as pressure, waves, and fields, that become apparent in the mutuality or because of the mutuality of physical and chemical entities are recognised as relative energy. Heat, sound, and electricity are also relative energy." — MVD, p. 40

"Without absolute energy, there is no basic impulsion in matter, and without matter, absolute energy is not revealed. Nature is eternally present in absolute energy." — MVD, p. 40

"The entire matter is saturated and contained within absolute energy, and the state and motion of the whole universe is within this absolute energy." — MVD, p. 45

Reading: *Absolute energy* \= *satta* itself, as the inherent, ever-present ground — inseparable from matter but not a force acting on it. *Relative energy* \= everything that shows up only *between* units: pressure, waves, fields, heat, sound, electricity. This is the clearest single sentence in the corpus for "which physical phenomena are unit–unit, and which are satta–unit": by MVD's own definition, **every named physical force (heat, sound, electricity, pressure, waves, fields) is relative energy — i.e., strictly inter-unit.** Satta's own energy is never itself named as one of the physical forces; it is what the physical forces are various manifestations of, at one remove.

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4. The saturation bond itself is described in force language

Here is the tension worth sitting with. Section 2 above establishes that *satta* does not act, has no pressure, no wave, no motion. But when the texts describe what saturation *does* to a unit, they switch into force vocabulary — repeatedly, not as a one-off metaphor.

“The essence of mutual recognition and regulation of units is their saturation and energisation in Omnipotence only. This submersion in Omnipotence is why every unit has space all around it. **Being soaked itself is forcefulness.** It becomes clear that units are soaked because Omnipotence is permeating… Therefore, a unit is seen to have regulation due to its being surrounded by Omnipotence, activeness due to its submersion in Omnipotence, and forcefulness due to its being soaked in Omnipotence.” — SB, p. 57

“**The force in a unit is essentially magnetic force, signifying its being soaked in Omnipotence.** Since the forcefulness is there through and through in the unit, the constancy of effort becomes self-evident.” — SB, p. 61

“**Saturation in uniform energy itself is forcefulness, forcefulness itself is basic impulsion, basic impulsion itself is activity, activity itself is effort-motion-result**, and effort-motion-result itself is development and its continuity.” — SB, p. 62

“Every one in existence, as a result of being the grandeur of saturation in Omnipotence, is proven [to have] forcefulness, **manifested in the form of magnetic forcefulness**. Hence, forcefulness is existentially inherent in each atomic particle.” — SB, p. 98

“It is only because formful existence remains saturated in formless existence (Omnipotence, Omnipresence) that **forcefulness is expressed**… every unit — molecules, atoms, and even the particles contained within atoms — remains regulated within a definite distance from one another, **which itself is formless existence**.” — SB, p. 79

Reading: the texts’ own resolution (implicit, not stated as a resolution) is that saturation is an *eternal endowment*, not an *event* — “forcefulness” names what is always-already true of a soaked unit, not a transmission that happens at a moment in time, the way one billiard ball strikes another. SB p. 62’s chain (saturation → forcefulness → basic impulsion → activity → effort-motion-result) is presented as one continuous unfolding “internal” to the bond, not a causal relay between separate things. Even so, the literal choice of “magnetic force” (चुम्बकीय बल) for a bond that is also called actionless is a real interpretive knot — see §15.

KD states the same endowment chain independently, in the context of how atom-particles recognise one another at all: “Every single thing, from the smallest to the biggest unit, is submerged, soaked, and surrounded in the omnipresent… **It is by virtue of being energy-endowed that force-endowment, and magnetic-force-endowment, is understood.** It is this force by which the particles of the subtle atom recognise one another in mutuality and participate in order.” — KD, p. 54 (our translation). The same passage grounds recognition-and-regulation (the two verbs SB p. 57 derives from saturation) in this endowment — so the SB and KD chains match link for link: saturation → energy-endowment → (magnetic) force-endowment → recognition → regulation/order.

SB p. 79 is also the clearest textual basis for “distance is satta”: the gap between any two particles or units is not empty in the sense of *nothing there* — it is formless existence itself. Space (a name the texts also give *satta* — see MVD p. 40’s list: Uniform Energy, Space, Knowledge, Consciousness, Absolute Energy) is not a container satta sits inside; it *is* satta, seen from the side of separation between bounded things. KD ch. 3.14 makes the same claim operational, extending it to measurement itself: “Whether it be far or near, every unit remains contained in the pervasive entity… **whatever we measure in the form of distance, amid mutuality there is only the pervasive entity**… between pole and pole we are measuring pervasiveness itself.” — KD, pp. 125–126 (KD-English 3.14). Every distance measurement is, for the darshan, a measurement *of satta* — the one place the unmeasurable ground is said to be what rulers in fact measure.

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5. The Five Forces table — physics mapped onto *jeevan*’s faculties

This is the most striking single artifact in the corpus for this question: SB contains an actual table pairing the four fundamental forces of physics, plus a fifth, against the five faculties of *jeevan*.

Insentient Powers — The Five Forces	*Jeevan* faculty	Sentient Power
Electromagnetic Force	*Mun*	Hope
Gravitational Force	*Vritti*	Thought
The Weak Interaction Force	*Chitta*	Visualisation
The Strong Interaction Force	*Buddhi*	Resolve
Mediative Force	*Atma*	Realisation
— SB, p. 85		

The surrounding text explains why the atom, specifically, is where all five are said to be visible:

“The study of existential measure concerns the atom as it exists in accordance with the destined orderliness of nature, endowed with generative, degenerative, and mediative powers. **It is only in the atom that all forces are seen to be fully operative.** In post-forms, namely molecules and molecular compositions, the mediative force is not apparent. For this reason, the study and teaching of all five forces become accessible only through the atom.” — SB, p. 84

“The mediative activity is inherently a manifestation of mediative force and power, **situated at the centre of the atom**. It continuously maintains control over the generative and degenerative forces… Insentient powers are subject to erosion, whereas sentient powers remain inexhaustible.” — SB, p. 84

Reading: the darshan takes over the standard four-force inventory of physics wholesale (electromagnetic, gravitational, weak, strong) and adds one the texts consider missing — a “mediative force” (*madhyastha bal*), located at the atomic nucleus, whose job is to regulate/normalise the other four ("generative and degenerative powers"). This fifth force is presented as *the* reason an insentient atom can, on developing, become a sentient one — mediative force is insentient nature's structural analogue of *atma*. It is worth being careful in the paper about what kind of claim this is: it is not proposing a fifth force for physics to detect: it is a structural analogy (insentient force : atom :: sentient faculty : jeevan), used to argue that jeevan's five-faculty architecture is not arbitrary but mirrors a five-fold structure the darshan already claims to see in insentient matter. Whether this is meant descriptively (physics literally has five forces once you count correctly) or analogically (a template shared across orders) is not disambiguated in the passage itself — flag as open in §15.

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6. Atomic structure: nucleus, orbits, and the absence of a numbered periodic table

This is the most direct material for "atomic number and orbital shells," and it turns out to be more developed than the Five Forces table alone suggests. The darshan gives every atom — insentient or sentient — the same generic architecture: a central particle-cluster (nucleus, *madhyansh*) surrounded by particles arranged in concentric orbits (*parivesh*, literally "surroundings," translated here as orbit/shell).

"Particles in every atom are situated either in the nucleus or in the surrounding orbits. The first orbit contains one or more particles continuously revolving around the nucleus; the same is the case with the second, third, fourth, and so on. Space pervades all around each atom as well as all around the particles within the atom's constitution." — MVD, p. 78

"The kind, state, and existential measure of atoms are determined by differences in the number of particles in their nucleus and orbiting particles." — MVD, p. 42

"The form and properties have changing nature, while the elements have perpetual nature, as they are found in all the four orders of nature. The basic element in the constitution of all large entities is the atom. **The constitution principle of atoms of all elements is the same. An atom consists of a nucleus and orbiting particles. All orders of nature arise from the activeness of particles in atoms and their position in the development progression.**" — MVD, p. 149

6.1 Growth rule: how orbits and mass build up together. AVD gives the clearest statement connecting orbital structure to weight (cross-reference §10):

"Constitution of the atom is in the form of the nucleus at the centre, and the dependent particles at well-defined distance... The dependent particles orbit around the nucleus. Thus, motion-paths (orbits) are the integral part of an atom. **More particles in an atom means more orbits also. As the number of particles in orbits increases, the number of particles in the centre (nucleus) also increases. This is how weight bondage appears in atoms.**" — AVD, p. 185

6.2 Orbital distance sets charge/regulation. AVD ties the same architecture to §8's charge discussion (the ± sense — see §8's terminology note), giving the nucleus an explicit regulatory role over the orbits:

"The regulation of generation and degeneration, or the regulation of excesses, is clearly the grandeur of the central particle of the atom. **Increase or decrease in the precise distance of orbital particles in an atom is referred to as generative or degenerative.** It is seen... that it always remains in equilibrium by the application of the strength and power of the nucleus." — AVD, p. 103

6.3 Differential activity of orbits, order by order. MVD states that which orbits are "active" differs systematically by order of nature — this is as close as the texts come to something like an electron-configuration rule:

"The particles only in the last orbit of all kinds of atoms within the material order are active through effort... Only the particles in the outermost and second-to-last orbits of insentient atoms are actively engaged in compounding activities... Particles in the fourth, third, and second orbits in the sentient atom (in animal order) are partially active... In the [knowledge order]... jeevan expresses its four and a half (4½) activities with activeness in the fourth, third, and second orbits." — MVD, pp. 78–79

6.4 The one case with a fully named orbital model: *jeevan* itself. When the "atom" in question is a constitutionally complete, sentient one, the nucleus and its four orbits get specific names and a fixed activity count — the clearest "shell diagram" anywhere in the corpus:

"The nucleus of the sentient unit (a constitutionally complete atom) is referred to as *atma*. The particles in its first orbit are referred to as *buddhi*, those in the second orbit are referred to as *chitta*, those in the third orbit are referred to as *vritti*, and those in the fourth orbit are referred to as *mun*." — MVD, p. 78

"*Atma*... is a part of jeevan which attains realisation in coexistence. I have seen it as the nucleus. It remains active as the nucleus of a constitutionally complete atom. All other activities of jeevan take place in its first, second, third and fourth orbits. There are only two activities in the nucleus of jeevan atom — realisation (*anubhav*) and authenticity (*pramanikta*)... *Buddhi*: two activities in the first orbit — enlightenment (*bodh*) and resolve (*sankalp*). *Chitta* : second orbit — contemplation (*chintan*) and visualisation (*chitran*). *Vritti*: third orbit — deliberation (*tulan*) and analysis (*vishleshan*). *Mun*: fourth orbit — taste (*asvadan*) and selection (*chayan*). These are all the ten activities of jeevan." — JV, p. 92

Reading: the darshan does have a real "shell model" — nucleus plus a variable number of concentric orbits, with particle count in each determining the atom's "kind" (*jaati*), its overall size/mass, and (via distance from the nucleus) its charge state. Growth is monotonic and coupled: more orbital particles requires a larger nucleus, and that pairing is explicitly what weight is (§10). Crucially, MVD p. 149 states this architecture is **one universal principle across all "elements" and all four orders** — material, bio, animal, knowledge-order atoms are not built from different kinds of parts; they differ only in how many particles occupy which orbits and how "active" (in the effort/ascending sense) each orbit is. The only place the orbit count is fixed at a specific number is *jeevan* — exactly four orbits plus a nucleus, ten activities total, each orbit given a specific psychological/cognitive name (*buddhi, chitta, vritti, mun*) rather than a numeric label. Ordinary (insentient) atoms are explicitly allowed more orbits ("first, second, third, fourth, and so on," MVD p. 78) — the texts do not cap or enumerate how many.

6.4a KD's quantitative additions: minimum atom, nucleus-to-orbit ratio, the four-orbit equilibrium, and the 108 species. KD ch. 3.1 (*Paramanu Sanrachna evam Anu Rachna*, pp. 53–57) restates the §6 architecture and then goes quantitatively further than SB/MVD/AVD anywhere (all quotes our translation):

"In the middle of the atom there is an arrangement of one or more particles, and around it, revolving in orbits, one or more particles. **With less than this, an atom does not obtain.**" — KD, p. 53 (a minimum-constitution axiom: 1 nucleus particle + 1 orbital particle)

"**However many particles are working in the orbits, at least that many are in the middle — more than that is also possible.**" — KD, p. 56 (a nucleus ≥ orbits particle-count inequality — sharper than AVD p. 185's qualitative growth rule)

"These orbits are one or more than one. **When orbits exceed four, the possibility of decreasing, and when fewer than four, the possibility of increasing, remains forever.**" — KD, p. 56 (four orbits as an equilibrium/attractor of the orbit count — see §15, where this bears directly on the jeevan-four-orbits question)

"It is from the difference in the number of working particles in atoms that difference in conduct is found. On the basis of such difference of conduct the species (*prajaati*) of the atom and the number of species are determined — **humans have so far claimed 108 species of atoms on this Earth, and remain hopeful of finding more species.**" — KD, p. 55 (the corpus's one acknowledgement of the chemical-element inventory — see the revised "what is not there" note below)

"**Weight depends on the number of nucleus particles (*madhyansh*) inherent in these atoms.** By virtue of this inherent weight one atom recognises another and they aggregate — this we call a molecule." — KD, p. 56 (weight keyed specifically to the *nucleus* count, not total count — a refinement of §10.3)

6.4b KD on particle transfer: excitation precedes displacement. KD p. 57 adds a process account absent from the other five texts — how atoms change their particle count at all:

"Changing of the number of particles, and as a result of the changing, is *pariṇām* (result)... The name of this activity is *prasthāpan–visthāpan* (absorption–displacement). **Before any particle is displaced from an atom, that atom is found to be excited (*āveshit*); it is only as a result of being excited that displacement (shedding) from an atom becomes possible. That particle merges into some other atom** — this is absorption. After displacement, the displaced atom is in natural-state motion; the absorbing atom, having absorbed the particle, is established in the dignity of natural-state motion." — KD, p. 57 (our translation)

Reading: this is the corpus's transmutation mechanism, and it knits §6, §8, and §9.2 into one loop: excitation (*āvesh* in its state sense, §8) is the *precondition* for particle emission; emitted particles are absorbed by other atoms; both donor and receiver then relax into natural-state motion; and since particle count fixes species (§6) and *pariṇām* names exactly this count-change, "result" — the third term of KD's own śram–gati–pariṇām principle — turns out to mean, at the material level, atomic transmutation. KD p. 117 defines all three terms of the triple at the atomic level: "In every atom, activity is in the form of effort, motion, result. **Effort is in itself the grandeur of the existent-state; motion is in itself the grandeur of spreading effect; result is found to be the grandeur of a different existent-state**" (KD-English 3.12). Radioactivity (§9.2's excited, overfull atoms shedding internally) is then a special case of *visthāpan*.

6.5 Above the atom: states of matter (solid, liquid, gaseous) and the elemental/compound/mixture taxonomy. The texts also carry a small but explicit bulk-matter taxonomy sitting on top of the atomic model:

"All matter exists in solid, liquid, or gaseous (air) forms and can be found in elemental, compound, or mixture states." — MVD, pp. 41–42

"Elemental state (*tatvik*): Composition of a homogeneous group of atoms is referred to as elemental." — MVD, p. 42

"Physical entities exist in solid and gaseous forms, while chemical substances manifest in solid, liquid, and gaseous states." — SB, p. 97

"The splendour of chemicals manifests in various forms: solid, liquid, and gaseous. For instance, water exists as ice in its solid form, as liquid water, and as steam in its gaseous form." — SB, p. 142

"By this stage, the solid substances take on their complete forms. Consequently, the grandeur of chemical substances emerges through solid, liquid, and gaseous processes. The formation of biological cells and prana sutra is one historic milestone from solidification processes of chemical entities." — SB, p. 145

"The insentient world, which we call material order and biological order, exists in solid, liquid, and gaseous states." — JV, p. 14

Reading: three states, not four — plasma never appears as a state of matter anywhere in the corpus (the word occurs once in SB, p. 204, as a translation of *rasa* in the seven bodily *dhatus*). MVD's "elemental = a homogeneous group of atoms" is the closest the corpus comes to defining a chemical element, and it is consistent with §6's picture: an element is a *population* of same-kind atoms (same particle count), not an entry in a numbered table. One quirk worth preserving: SB p. 97 gives *physical* entities only solid and gaseous forms, reserving the full solid–liquid–gas triple for *chemical* substances — liquidity is apparently treated as evidence of chemical (rasa-yielding) activity, which coheres with SB p. 145's claim that biological cells emerge from "solidification processes of chemical entities."

What is not there: none of the six texts assign a specific particle count to a specific named element (hydrogen, oxygen, iron, gold, etc. all appear only as generic illustrations of "-ness," e.g. “ironness,” never with a stated number of nucleus/orbit particles) — so there is no periodic table proper, no per-element atomic-number assignments, and no electron/proton/neutron vocabulary anywhere in this corpus (verified for KD as well: no Devanagari loan-words for electron, proton, or neutron occur). KD p. 55 does, however, explicitly acknowledge the element inventory *as a human claim* — “humans have so far claimed 108 species of atoms” — so the darshan is aware of, and accepts the reality behind, a discrete numbered series of atomic kinds; what it declines to reproduce is the table’s contents. What exists instead is the growth law (more particles → more orbits → bigger nucleus → more weight → different “kind”), now sharpened by KD’s constraints (§6.4a: minimum 1+1 constitution; nucleus ≥ orbit count; four-orbit equilibrium) — still without the per-element numbers that would let you look up carbon’s structure the way you would in a chemistry textbook.

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7. Motion: rotational, orbital, vibrational — and “excited” (SB: “charged”) vs. “natural-state” motion

“Every unit exhibits motion in the rotational, orbital and vibrational form… Vibrational and orbital motion is present in every unit… As [an atom] develops, its vibrational motion keeps on increasing. Alongside this, its stability also [increases]… the moment constitutional completeness is achieved, that atom’s vibrational motion [reaches its natural ceiling].” — SB, pp. 53, 59

“Motion, in every unit, is [categorised] as rotational, orbital, and vibrational… this motion, together with environmental pressure, results in displacement. ‘Place’ means equal energy — or space and energy-fullness — because every unit is a bounded body from six directions… Every unit exists, in its mutuality, either in natural-state motion or in charged motion. The primary cause of charged motion is the pressure arising from the combination of naturalness and environment.” — SB, p. 59

Reading: the darshan’s basic kinematics is: every unit (atomic or otherwise) always has rotational + orbital + vibrational motion, intrinsically (this is closer to “zero-point” activity than to Newtonian rest). What varies is whether that motion stays in its *natural state* (*swabhav gati*) or becomes *excited motion* (*aveshit gati* — SB’s “charged motion”) under pressure from environment/naturalness — i.e., external interference is what turns ordinary intrinsic motion into excited, disturbed motion. This maps fairly directly onto §8 below (excitation), and the “orbital” component maps onto §6’s shell structure.

7.1 KD’s force/power distinction — and its indictment of physics. KD ch. 3.11 (“Force–Power (State–Motion)”) formalises a distinction the other texts leave implicit, and aims it at modern physics directly (our translations):

“From the viewpoint of Madhyasth Darshan it is learned that **in state (*sthiti*), force (*bal*) is manifest; in motion (*gati*), power (*shakti*) is manifest.**” — KD, p. 102

“In the modern-science era the attempt has been to recognise pressure as force, and motion, wave, and flow as power — electrical waves and magnetic force likewise… In all of this it is found that **the fact of mediation (*madhyasthīyakaran*) has been denied. All these studies are based on excited (*āveshit*) motion alone. When [a thing] is in natural-state motion, where do force and power go? — that question inevitably arises.**" — KD, p. 102

"Every unit is **force-endowed in state, and evidences power in motion.**" — KD, p. 103

Reading: *bal* (force) is what a unit *has* by virtue of being (its endowed, standing side — §4's saturation-forcefulness); *shakti* (power) is what shows up when it *moves/acts* (the expressed, relational side — §3's relative energy). The pair maps onto sthiti/gati exactly as absolute/relative energy maps onto satta/mutuality. KD's critique sentence is the corpus's most precise indictment of physics in physics' own terms: its instruments and formalisms only ever register *excited* motion (disturbances, differences, §8 — SB's "charged motion," *āveshit gati*), so its inventory of forces omits whatever holds in the *natural* state — which is where the darshan locates the mediative. Note also KD p. 75 closes the definitional circle from the heat side: "**normal temperature means being in natural-state motion**" — the unexcited state is simultaneously the thermally ordinary one.

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8. Charge and excitation (*āvesh*): generative/degenerative powers, the nucleus, and "hidden energy"

"Activeness itself is recognised as generative, degenerative, and mediative powers. **Generative and degenerative powers are observed as the charge in mutuality**, while the function of mediative activity is to normalise this charge. This mediative power is commonly known as **the hidden energy**." — SB, p. 70

"The physicochemical atom's positive or negative charge refers to the increase or decrease [of peripheral particles' distance from the mid-particle]... Positive and negative charges [only] prove relative — that is what 'more' and 'less' mean. Neither 'more' nor 'less' is complete, and for that reason neither is stable. The mediative power therefore works continuously to normalise the negative and positive charges (powers), because the goal of mediative power (activity) is completeness." — SB, p. 70

(Same distance-from-nucleus definition of generative/degenerative excitation appears independently in AVD, p. 103 — see §6.2.)

"That is, in the regulated state, there is no charge; this itself is the natural state. In the charged state, the possibility of decline naturally remains inherent. The decline itself is the evidence of charge." — SB, p. 79

"...degenerative charge is always a manifestation of a problem." — SB, p. 88

A terminology note first: the Hindi root *āvesh/āveshit* is used **contextually** in the texts, and the translation must follow the context. **(a) The ± sense:** where the text speaks of the positive–negative (धन-ऋणात्मक) imbalance of atoms — defined by orbital particles' distance from the nucleus — *āvesh* is **charge**, in the electric-charge sense (SB pp. 70–71: "the physicochemical atom's positive or negative charge refers to the increase or decrease in distance of the orbiting particles from the nucleus"; p. 250: "possible to measure positive and negative charges"). **(b) The state sense:** where the text contrasts a unit's neutral/natural state with a disturbed one — *āveshit gati* vs *swabhav gati* (§7), an overfull atom's condition before expelling particles (SB pp. 72, 99, 259), a heated body's internal condition (SB p. 252) — *āveshit* is **excited**, and *āvesh* **excitation**. SB's published English uses "charge/charged" for both senses, but glosses the state sense itself at p. 257: "entities are in a natural state or in an **excited (charged)** state." MD-Mapping encodes both readings (आवेश \= "agitation, charge, excitement"; आवेशित गति \= "agitated state, excited state"), and its Hindi definition of *āvesh* is explicitly value-laden — "deluded conduct-tendency; interference, motion toward decline." This note and the KD-English translations apply the contextual rule in their own voice; verbatim SB quotations keep the published wording. This settles, from the vocabulary side, what this section's reading argues from the content side: "charge" in the darshan names an imbalance-state, not an electrical quantity — and MD-Mapping's "motion toward decline" definition is precisely SB p. 79's "in the charged state, the possibility of decline naturally remains inherent."

KD corroborates and extends the excitation doctrine on three independent pages (our translations): heat-side — "an object that has become more excited is always heated; that is, its temperature rises" (KD, p. 74; cf. SB p. 252's heat \= excitation, §9); process-side — an atom must already be excited before it can shed a particle (KD, p. 57; §6.4b); and pressure-side — "**pressure (*dabāv*) is the compulsions received for generative-degenerative (*sam–viṣam*) excitation, as distinct from natural-state motion**" (KD, p. 118), which folds SB p. 49's pressure definition (§2) explicitly into excitation vocabulary.

KD p. 103 also gives the nucleus's §6.2-style regulation an explicitly *two-directional* dynamical form: "If, within that definite good distance, the distance increases, then the mediating activity working as the nucleus (*madhyansh*) applies its force and calls the particles near. If a particle begins to approach the nucleus, then too the nucleus, applying its force, fixes the particle at the good distance. **This is the control activity occurring in the atom.**" (our translation) — i.e., the mediative acts as a restoring influence toward a set-point separation, against displacement in *either* direction. See §14.3 for what this does to the mediative-as-primitive claim.

KD p. 77 (KD-English 3.7) further embeds excitation in a universal directional chain: "**every excitation is understood to be working toward… natural-state motion; natural-state motion toward development progression; development progression toward evidencing development; development toward evidencing awakening progression; awakening progression toward evidencing awakening**, and toward evidencing the continuity of awakening." De-excitation is thus not merely local relaxation but the first step of a cosmic monotone — the darshan's arrow of change (taken up in §14.3).

Reading: the two senses of *āvesh* are systematically linked in SB's doctrine. In the ± sense, **charge** names the atom's generative/degenerative imbalance — generative power in excess of degenerative, or vice versa, measured by orbital distance from the mediative reference point at the nucleus. Positive and negative charge are explicitly called *relative* — a matter of "more or less" around a mean, never an absolute or stable quantity — and mediative force's job is continuously pulling any excess back toward balance. An atom carrying such an imbalance *is thereby* in the **excited state** — the state sense: the natural, unexcited state is definitionally the regulated (charge-free) one, and being excited is already partway toward decline. So the ontology runs: ± charge (imbalance) → excited state (disturbed condition) → decline, unless the mediative normalises it. This is a value-laden physics: excitation is functionally close to what SB elsewhere calls "problem," and its absence is close to what it calls "resolution" (*samadhan*) — consistent with the book's title.

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9. Heat, magnetism, electricity, radiation

"Radiance and imaging are not forms of motion or pressure. Nevertheless, the intrinsic properties inherent in every entity are found to exert influence. For example, **heat, magnetism, electricity, and weight influence one another**. The field of influence itself is part of their grandeur." — SB, p. 252

"On this very basis, heat, magnetism and electrical effects are projected… **Fundamentally, heat equals the charge of a unit.** The greater the internal charge within a planetary body, the more it tends to transform into atomic species of progressively lower numerical constitution." — SB, p. 252

"The coordination and harmony in [biological cells'] mutuality is manifested through pulsation. Through contraction and expansion, they operate respectively as pressure and wave… [and] the entire biological-order unit is proven to be **electrically receptive**." — SB, p. 83

On the modern-physics method of studying atoms by breaking them (particle-collision / fission-type methods), SB is explicitly critical:

"In this approach, the pressure method is regarded as the most effective means of achieving fragmentation or destruction. For instance, **applying magnetic force to an atom generates a charge within it, leading to the disintegration of its components**… This reveals that the force responsible for such distortion is external to the atom itself — specifically, **electromagnetic force** — which interacts with and alters the atom's natural state." — SB, p. 128

"…it was through natural correlation that humans, through imaginativeness and freedom of action, came to recognise the natural phenomenon of electricity… [and], from prior events, had already identified magnetic elements, leading to the establishment of a tradition of studying magnetism." — SB, p. 128

Reading: heat, magnetism, electricity, and weight are grouped together explicitly as one family — properties that "influence one another" through a "field of influence," and are explicitly *not* forms of motion or pressure in themselves (they are effects of the underlying excitation/pressure dynamics, not identical to them). Heat is given a specific reductive definition: heat \= excitation (SB: "heat equals the charge of a unit"). SB's critique of experimental particle physics (p. 128) is philosophical, not technical: it reads magnetic/electromagnetic manipulation of atoms as an *externally imposed* distortion, and contrasts this with the darshan's preferred method of studying wholes without destroying them (a recurring methodological theme — see also §6.4.2 of *The Ontology of Coexistence* on evidence standards).

9.1 The Sun, excess heat, and the "law of complementariness." Directly following the heat \= excitation passage, SB gives its one worked cosmological-scale application of the excitation doctrine:

"A heated object (e.g. the Sun) reaches its natural state according to development progression. Until such a state is established, the entire environment — including the Earth and other planets of comparatively lower temperature — **distributes among itself the excess heat projected from the Sun**. As a result, the possibility remains for the Sun to become free from excess heat through **the law of complementariness**." — SB, p. 252

9.2 Radioactivity: "overfull" atoms and internally deployed heat. SB has an explicit account of radioactivity and nuclear explosion, entirely in the vocabulary of §6 (particle counts) and §8 (excitation and mediative regulation):

"**Atoms that have become overfull are found to be radioactive by nature.** The very definition of radioactive substances is that, within radioactive atoms, intrinsic atomic heat or fire begins to deploy internally. From this principle, it becomes clear that in an excited or interfered state, the pressure of internalised heat may increase to such an extent that nuclear material reaches an explosive condition. On this basis, it now appears reasonable to infer that **in the Sun, the maximum number of overfull atoms have become charged, resulting in continuous nuclear explosions**." — SB, p. 259

The preceding page attributes human-triggered explosions to interference with the nucleus's mediative regulation (§5, §8): "substances, which are situated beneath the Earth's surface and function there according to the law of complementariness, are being extracted… As a result of this interference, the naturally mediative nuclear activity — mediative force and mediative power — becomes incapable of balancing the charged components, leading to explosions." — SB, p. 258

Reading: this closes a loop the earlier sections leave open. §6's growth rule said more particles → more orbits → bigger nucleus; here we get the failure mode of that progression: an atom can become *overfull*, at which point the mediative force can no longer normalise its generative/degenerative excitations and the excess deploys internally as heat — radioactivity. The Sun is then read not as a fusion reactor but as a vast population of overfull, excited atoms in continuous explosion, whose excess heat the cooler planets absorb (§9.1) until the Sun too "reaches its natural state." Radioactive decay, solar output, and nuclear weapons are thus one phenomenon at three scales, and all three are excited states — i.e., in SB's value-laden register (§8), problems awaiting resolution. Note the direction of explanation is the reverse of standard astrophysics: the Sun is cooling *toward* its natural state, not burning through fuel.

9.3 The critique extended: quantum mechanics, relativity, and fission. AVD broadens SB p. 128's methodological complaint into a survey of physics' own self-descriptions:

"Specialisation has come to a purposeless dead end in all disciplines. For example, **quantum mechanics is facing the dead end of instability-uncertainty**." — AVD, pp. 63–64

"Specialists of the theory of relativity are using their imagination to explain the phenomenon not completely understood so far... They start their discussion with kinetic energy; for example, conversion of magnetic field into electric current, which may be used as work energy. And then somewhere they start talking about waves. **Experiments are still being conducted to conclude whether to categorise these waves as made up of particles or not... Uncertainty still looms regarding whether waves are made up of particles or not.**" — AVD, p. 64 (the translator inserts a bracketed "From internet:" gloss on wave–particle duality at this point — translator's addition, not source text)

"The concept of splitting was arrived at by a mathematical approach. **Nuclear fission or splitting of the atom was discovered for making increasingly more powerful nuclear bombs.**" — JV, p. 119

Reading: the corpus's engagement with named modern physics is thus wider than SB p. 128 alone: Big Bang cosmology (§2.1), quantum mechanics and wave–particle duality, relativity, and fission are all addressed by name, and always with the same two-pronged objection — epistemic (uncertainty/instability as a *dead end*, not a finding) and moral (fragmentation-by-force as destruction, fission as bomb-making). Nowhere is the mathematics engaged on its own terms; the objection is to the method and its motive, consistent with SB's "mathematics is more than the eyes and less than understanding" (p. 87).

9.4 KD's heat mechanics, waves, electricity, and celestial order. KD ch. 3.7 ("Heat and the Earth's Balance") and ch. 3.13 ("Pressure, Flow, Wave, Electromagnetic Force") contain the corpus's most worked-out phenomenal physics (all quotes our translation):

Heat and its flow. "Where there is heat, if around it there is lesser heat, only then does the greater heat get reflected (*parāvartit*) toward the lesser." — KD, p. 75. Heat spreads by molecules near the source becoming hot and expanding, passing heat molecule-to-molecule until everything "arrives at normal temperature — and normal temperature means being in natural-state motion" (KD, p. 75). Solids expand when heat is introduced, becoming liquid, then rarefied (KD, p. 74); heated iron "keeps reflecting heat until it has cooled, distributing its heat among all associated substances" (KD, p. 74). This is a hot-to-cold flow law (a second-law-like asymmetry) stated in the vocabulary of *parāvartan* (reflection — ch. 3.12's own category), with thermal equilibrium defined as return to natural-state motion.

Waves, by state of matter. "In solid, liquid, and rarefied [matter], waves occur in different ways: in solid objects, waves are in the form of vibrational motion; in liquid, waves are the reaction produced by contact of air or objects; **in rarefied substance, sound (śabda), heat (tāp), electricity (vidyut), ray (kiraṇ), and radiation (vikiraṇ) are evidenced in wave form.**" — KD, p. 118. Sound arises "from the friction or collision of one or more objects" (two stones rubbed, metals struck, leaves in wind — KD, p. 118); word-waves travel by pressure imposed on the molecular groups of the atmosphere, "transmitted molecule to molecule… reaching far — radio and television waves are held to be word and sound propagated electricity-borne" (KD, p. 119).

Electricity from magnetism. "**Electric wave: this is the flow produced by the fragmentation (*vikhaṇḍan*) of magnetic current.** The small fragments into which magnetic current is broken are transmitted through some medium — that is, they run. As for the medium: the whole of nature is found to be the bearer-carrier of electric current. The whole of nature is electricity-receptive, at minimum or at maximum; this more-or-less is settled by pressure, flow, and quantity." — KD, p. 118. Lightning is the electromagnetic wave seen naturally; thunder is the sound of dense clouds approaching one another from their "good distance"; when they near each other and connect with the Earth, "electric flow runs of itself" (KD, p. 119); the dynamo — "the fragmentation-of-magnetic-current method… rotated with a bicycle wheel" — is how humans obtained light from it (KD, p. 119).

Celestial order. The Sun's heat "is being reflected equally in all directions"; the Earth-facing side becomes hot, which is why sunrise/sunset and the day's temperature curve occur (KD, p. 76). The Earth's rotation presents its faces to the Sun; its revolution around the Sun proceeds at a maintained "good distance" (*acchī dūrī*) along with all the planet-spheres of the solar array, "and **this circling path, being not fully circular but elliptic (aṇḍākār)** — this is the tendency — has as the chief intent of that tendency the Earth's self-organised tendency of seasonal change (*ṛtu parivartan*)." — KD, p. 76.

Reading: three things stand out. First, the *direction* of KD's explanations: heat flows hot→cold not as statistics but as mutual "recognition–fulfilment" between substances (the melting iron "accepts" deployed heat — KD p. 74 explicitly calls this acceptance *recognition*, extending §4's recognition doctrine into thermodynamics). Second, magnetism is treated as *prior to* electricity (electricity \= fragmented magnetic current) — the exact inverse of the electromagnetic hierarchy, but consistent with the corpus's decision to make "magnetic forcefulness" the fundamental endowment (§4). Third, the celestial mechanics is teleological: elliptical orbits and maintained "good distance" are not consequences of gravitation but expressions of order whose *purpose* is seasons, and ultimately the development-progression toward jagriti on Earth (KD, p. 76). Note the ellipse itself is correctly asserted — what differs from Kepler is the direction of explanation, the same pattern as §9.2's Sun.

Three additions from the full KD-English translation of ch. 3.7: (a) the Sun is described outright as "in the ultimate condition of decline," destined "in the course of time" to enter development progression (KD p. 77) — the strongest single confirmation of §9.2's cooling-Sun reading; (b) the hotter-to-cooler heat transfer among bodies is generalised to astronomy: planet-orbs of maximum heat, "such as there is in the sun," have their heat absorbed "into all those planet-orbs facing them in which less heat remains" (KD pp. 77–78 — KD's version of SB p. 252's law of complementariness, §9.1); (c) the Earth's **pole-to-pole magnetic current** is said to be what keeps the planet solid — if overheating deflects it, the Earth would "begin to scatter" (KD p. 81) — magnetism again given the structurally prior, binding role (cf.

§4). One caution for the paper: KD ch. 3.12 shows *parāvartan–pratyāvartan* (reflection–projection, per the project's settled rendering) to be a fully general category — teaching/study, evidence-giving, and heat transfer are all its species — so ch. 3.7's thermal "reflection" should not be read as a specifically optical or thermodynamic technical term.

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10. Mass and weight: the most concrete explanatory chain in the corpus

This is the best-developed physics-adjacent explanation in the primary texts, and it directly answers "how do atoms get weight/mass."

10.1 The ground state is weightless. JV, in Q\&A form, states this as an axiom:

"According to materialism, where units exist, there is no space, meaning units displace space. Therefore, materialism considers units mightier (heavier) than space. However, in reality… all units exist in a state of **zero-attraction**. Earth, Sun, solar systems, and galaxies are all in a state of zero-attraction. Zero-attraction is the absolute state, indicating that **there is no weight in the coexistence of Omnipotence and units. Weight becomes evident only in the interaction of units.** For example, weight is evident in the mutuality of two atoms due to their tendency to form molecules. Similarly, weight is evident among molecules due to their tendency to form definite compositions. **This manifestation of weight in the mutuality of units is commonly called the gravitational force.**" — JV, pp. 149–150

10.2 Three attraction-states, and a derivation chain for weight. MVD:

"Every planet is in a state of zero attraction in space, and the position of each planet is determined by its motion in space. Units in the state of **positive-attraction** are attracted to something, units in the state of **negative-attraction** attract something, while units in the **zero-attraction** state are independent." — MVD, p. 45

"**The weight depends on attraction; attraction depends on mutuality; mutuality depends on smallness or bigness; smallness or bigness depends on composition and relative energies; composition and relative energies depend on activity; activity depends on matter; and matter's development or decline depends on the right-use or misuse of relative energy.**" — MVD, p. 45

10.3 Atomic mass \= particle count. (Same principle as §6's orbit/nucleus growth rule, restated directly in terms of mass:)

"**The mass and lineage (or category) of atoms are determined by the number of their constituent particles.** Their further development is clearly understood based on the principle of coexistence. For instance, the formation of an atom from two constituent particles [is the simplest case]." — SB, p. 182

"In this way, the atom is established as **the basic quantum**. Since molecules and molecular forms arise from atoms, understanding atoms as the fundamental unit of measurement naturally provides the key to comprehending the quantity of all bodies." — SB, p. 86

"More particles in an atom means more orbits also. As the number of particles in orbits increases, the number of particles in the centre (nucleus) also increases. This is how weight bondage appears in atoms." — AVD, p. 185 (see §6.1)

10.4 Sentience is explicitly freedom from weight.

"Every sentient unit i.e. *jeevan* is **free from weight-bondage and molecular-bondage**." — SB, p. 114

"A constitutionally complete atom is **free from molecular and gravitational bonds**, and this itself is its grandeur." — JV, p. 150

10.5 KD: weight as evidencing coexistence, and "gravitation" renamed. KD ch. 3.11 gives the JV/MVD doctrine of §10.1–10.2 its most explicit form, including the one place in the corpus where the word *gurutvākarṣaṇ* (gravitation) is named and re-defined (our translations):

"**The whole of weight is the arrangement (*vinyās*) occurring in the course of evidencing coexistence with the Earth** — that is the principle. In this way weight is found to be animated by coexistence." — KD, p. 103

"Likewise the solid object that falls from above — that is the arrangement occurring with the Earth, which happens in the effect of that object's evidencing its coexistence with the Earth. **To this arrangement the name 'gravitation' has been given.** In reality it is only the arrangement of a solid evidencing coexistence with a solid… **From 'gravitation' runs the conception of small and big. Seen as coexistence, each one is complete together with its environment. In that, there simply is no small or big.**" — KD, p. 104

"The arrangement of this tendency is found to remain bounded (*maryādit*) up to this Earth's environment. This Earth too is complete together with its own environment." — KD, p. 104

Molecule formation is put on the same footing: atoms join "on the basis of the mid-particles, mutual attraction-force operating on the basis of magnetic-force-endowment," molecules are naturally weight-endowed, and "because of this weight, the tendency persists for one molecule to join another while staying at the definite good distance" (KD, p. 103) — one mechanism from atomic bonding up to falling bodies.

Reading, pulled together: weight/gravitation is, by explicit textual statement, not a property of matter in isolation and not something *satta* imparts. It is a name for a *tendency units display toward each other* — specifically, a tendency to combine into larger compositions (atoms→molecules→bodies) — that only becomes "evident" once more than one unit is in mutual relation. The dependency chain in §10.2 traces weight backward through attraction → mutuality → relative size ("smallness/bigness" — note *laghuta/guruta*, the same root that gives *gurutvakarshan*, the ordinary Hindi word for gravitation, so the etymology is doing real work here) → composition/relative-energy → activity → matter itself, terminating in "right-use or misuse of relative energy" — i.e., ultimately a normative rather than a purely mechanical vocabulary. Atomic mass proper is simpler and more concrete: it is just the count of constituent particles in an atom (nucleus + orbits, §6), which is also what fixes which "species" (element-analogue) the atom belongs to. The clean, repeated corollary is that a *jeevan* atom (constitutionally complete, sentient) is explicitly exempted from both molecular bonding and gravitational/weight bonding — weight is presented as strictly a feature of insentient nature's combinatorial tendency, one of the things a unit "graduates out of" at the sentience threshold. KD's contribution (§10.5) is to make the reversal of standard gravitation fully explicit: falling is not attraction by a larger mass but a body's *evidencing of coexistence* with the Earth; the small/big asymmetry that makes Newtonian gravitation asymmetric is dismissed as an artifact of the "relative method" (each thing being "complete with its environment," nothing is small or big); and — most testably — the tendency is *bounded to the Earth's own environment* rather than universal. That last clause puts KD in direct, statable conflict with universal gravitation (flagged in §15).

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11. Indestructibility: nothing is created, nothing is annihilated

This theme is stated independently in all five texts and functions as the darshan's analogue of a conservation law — though what is conserved is *units*, not mass or energy as quantities.

"Existence implies beingness and indestructibility." — SB, p. 49

"Since nature is saturated in Omnipotence, every unit is endowed with the dharma of existence, evidenced by the **indestructibility of units. All that occurs is merely change and development.** Dharma means that from which one cannot be separated. Since existence is coexistence, this (indestructibility) itself is the ultimate dharma." — SB, p. 50

"Even after fragmenting an atomic particle into countless parts, every fragment is still called 'one'. This demonstrates that **annihilation or obliteration of anything is impossible**… **In existence, there is no entity that 'comes into being', nor is there anything that 'ceases to be'.** These processes do not occur in reality." — SB, p. 103

"From ancient times till now, all the research, experiments and conjectures regarding partitioning and divisions can be summarised as — **nothing can be annihilated by progressive fragmentation.**" — AVD, pp. 69–70

"Just because an entity has decomposed does not mean annihilation or destruction of its respective constituents… **existence neither decreases nor does it increase**." — AVD, p. 245 (restated at p. 275: "Existence neither decreases nor does it increase, it is therefore free from profit & loss, and thus indestructible")

"To whatever extent we may subdivide them, it will only increase the number of units (particles), without anything getting annihilated." — AVD, pp. 247–248

"Every entity in its fundamental form is indestructible, no matter how many changes or results occur. In this way, the stability of the entire existence becomes clear by itself." — JVD, p. 82

"Physical entities continue to exist, these are never annihilated. The physical entities only manifest as chemical entities, which is called chemical resonance (*rasaynik urmi*)." — JV, p.

"There is no annihilation or absence of matter, *mun*, *vritti*, *chitta*, *buddhi*, and *atma*. They exist only in varying degrees of development and awakening." — MVD, p. 299

Reading: the corpus's conservation principle is *countability*, not quantity: divide a unit as many times as you like and each fragment is still "one" — still bounded, still soaked in satta (cross-reference SB p. 57 in §4: "No matter how many times we divide a unit, each fragment remains submerged, surrounded, and soaked… a unit cannot be annihilated"). What physics tracks as conservation of mass-energy the darshan replaces with a stronger, more ontological claim: composition and decomposition are the only real processes; creation *ex nihilo* and annihilation are not rare or difficult but strictly meaningless. This is the positive counterpart of §2.1's Big Bang rejection — no origin event is needed because "coming into being" is not something that happens. Note the asymmetry with physics: pair annihilation (particle + antiparticle → photons) would presumably count for the darshan as transformation of units, not annihilation, but no text addresses anything like antimatter, so the reconciliation is untested. MVD p. 299 extends the same principle across the sentient side — jeevan's faculties are as indestructible as particles, differing "only in degrees of development and awakening."

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12. Form — shape, volume, density — and the limits of measurement

The texts define every unit's *form* (*roop*) by a fixed three-dimensional triple, and then make an explicit claim about which of the darshan's quantities are measurable at all.

"Form is understood in terms of shape, volume, and density." — SB, p. 65 (repeated as the first of the four dimensions of atomic bodies at SB p. 86: "Form: as shape, volume, and density"; and as a definition at MVD, p. 50: "Form in all four orders is determined by differentiation of shape, volume, and density")

"An object in itself is in the form of shape, volume, and density. **This is the very framework of a unit, defining its boundary.** Every unit is bounded on all sides. Omnipotence is seen on all sides of every unit." — SB, p. 249

"Compositions made of atoms of a particular kind may vary in size, shape, volume, or density, but their essence remains constant… the intrinsic properties and nature of objects — expressed as their 'essence' — **do not increase or decrease despite variations in measurement, weight, or volume**." — SB, p. 119

"**Shape, volume, and density can be measured; all properties cannot be measured. It has become possible to measure positive and negative charges.** It is known that at the root of natural state and mediativeness lies the mediative power. **This mediative power has not been measurable.**" — SB, p. 250

"The volume of an object is always more than what the shape projects on the eye, and the density even more than the shape & volume. In this way, whatever we see with our eyes, its entirety is not captured with the eyes." — AVD, p. 199

"The shape, volume and density of jeevan, a sentient unit, can be understood in the context that it is also only an atom." — AVD, p. 185

"Powerful lenses that show lakhs and crores of times larger, electronic lenses for showing billions and trillions of times, have come into use. After all of that, what happens is in the eye: **instrument-evidence (*yantra pramāṇ*) cannot exceed the eyes.** Make the eye a billion times more effective — only that much meaning comes out." — KD, p. 133 (our translation)

"Even if, by the mechanical method, we do direction- and time-related activity, in that too all the activities… remain related to the human. Hence, **at the root of instrument evidence, the human alone is the evidence.**" — KD, p. 130 (KD-English 3.14); ch. 3.8 presses the same point against science's self-description: "the instrument became the principal evidence… responsibility, instead of being on the human, was thrust upon instruments… With instruments no reasoning is possible at all, because what an instrument does, it keeps doing again and again" — KD, p. 86 (KD-English 3.8)

Reading: density is the third coordinate of unit-hood itself — form *is* shape-volume-density, and that triple is what bounds a unit off from the satta surrounding it (tying directly to §4's "distance is satta"). SB p. 250 then draws the corpus's clearest line between what measurement can and cannot reach, and it maps exactly onto the §5/§8 force inventory: form dimensions — measurable; positive/negative (generative/degenerative) charges — measurable, and the texts credit physics with having measured them; **mediative power — not measurable**. This is the darshan's own explanation of why the fifth force of SB p. 85's table is absent from physics' inventory: instrument-based method can only register what varies (charges, "more and less," §8), and the mediative is definitionally the non-varying reference point of the variation. Essence-invariance (SB p. 119: "ironness" constant under any change of size, weight, or volume) plays the role that intensive properties play in physics, but is extended into a metaphysical claim. AVD p. 199 adds an epistemic ordering — eyes capture part of shape, less of volume, less again of density — grounding the recurring argument that perception, and by extension instrumentation, systematically under-samples even measurable form. KD p. 133 supplies the missing premise connecting the two: instruments are *amplified eyes* and nothing more ("instrument-evidence cannot exceed the eyes"), so AVD's limits on the eye transfer to every microscope and detector — which is why, for the darshan, no instrument could ever register the mediative (SB p. 250) and understanding (*samajh*) rather than magnification is the required instrument.

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13. Time (*kaal*) — brief cross-reference

Time is treated at length in the existing Nature of Time study; this note only pulls the handful of quotes most relevant to the satta–unit physical-interaction question, for completeness.

"Time (*kaal*): Duration of an activity is referred to as time." — MVD, p. 34

KD ch. 3.15 opens with the identical definition and adds a critique of clock-time (our translation): "The definition of time as **'the duration of activity'** is clear… On the basis of one cycle of a continuously recurring activity, time came to be recognised — the Earth in its one rotational motion completes day-and-night over the whole Earth, and this was named the day; this duration was divided into 24 parts, then 60… **In this manner we went on dividing time, and forgot the activity. Time became primary, the object was forgotten. From all the decisions taken on the basis of this mistake, problems were born.** … Time is ever-present (*nitya vartmān*); the present, as reality, is coexistence itself; it is to the present that we have given the name *kaal*." — KD, p. 134

Coexistence is *trikaalabadh* — a timeless truth valid across all three times. — SB, p. 48

"The difference between absolute energy and matter is that absolute energy is present in every time, place, and object. No place or object has been found or proven to be devoid of absolute energy. Conversely, there are places devoid of matter, though such a time is not proven." — MVD, p. 41

Reading: time is defined strictly as a property of unit-*activity* (*kriya*), exactly parallel to how §2–§3 above define pressure, force, and weight as unit-mutuality phenomena. *Satta* is *kriya-shunya* (actionless) and therefore, by the same logic, timeless — not because it exists "before" or "outside" time in a cosmological sense, but because time is definitionally absent wherever activity is absent. This is structurally the same move the texts make for force and weight: locate the physical-sounding predicate at the unit level, deny it of the ground, by definition rather than by discovery.

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14. Synthesis: the structure of the darshan's physics

Read section by section, the corpus can look like a list of unconnected physics opinions. Read together, §2–§13 are one construction applied uniformly. This section states the construction once (14.1), gives it a formal reading that organises it more tightly than paraphrase can (14.2), isolates the single claim in it that ordinary physics does not already hold in some form (14.3), and closes with the reference map (14.4).

14.1 The repeated move. Across pressure and cosmogony (§2), force (§3–4), atomic structure and states of matter (§6), charge/excitation (§8), heat/magnetism/electricity/radioactivity (§9), weight (§10), indestructibility (§11), form/density (§12), and time (§13), the texts do the same three things every time:

1. **Deny the predicate of *satta* directly and explicitly** — actionless, pressureless, wave-free, weightless (zero-attraction), timeless, non-transforming.
2. **Locate the predicate in unit-to-unit mutuality instead** — pressure from environmental influence; charge from generative/degenerative imbalance around a mediating centre; weight from units' tendency to combine; time from activity's duration.
3. **Name the bond between satta and units with force-like vocabulary anyway** — "forcefulness," "magnetic forcefulness," "energisation" — but frame it as an eternal

endowment/unfolding, not an event, force-transmission, or efficient cause.

Steps 1 and 2 are repeated independently across SB, MVD, JV, and KD, and compress into a single axiom, which is the corpus's settled position: **every physical predicate is defined only on relations between units; no physical predicate is defined on a unit in isolation, and none on the ground.** MVD p. 40's relative-energy definition (§3) is this axiom in the texts' own words — heat, sound, electricity, pressure, waves, fields are all "power which doesn't become apparent without the mutuality of units" — and KD p. 102's *bal/shakti* pair (§7.1) is its dynamical form: force is what a unit is endowed with in state, power is what appears only in motion, i.e., in relation. Step 3 is the one place the system's vocabulary pulls against itself (§15, first bullet) — though 14.2 suggests the tension is terminological rather than structural.

14.2 A formal reading: units as objects, mutuality as morphisms, satta as the unit object. *This subsection is our interpretive frame, not the texts'. It is included because it resolves two of §15's open tensions and states the darshan's position in vocabulary a physics-literate reader already has.*

Take units (*ikai*) as the objects of a category and mutualities as its morphisms. The axiom of 14.1 then reads: **observables are properties of morphisms; objects carry no intrinsic physical predicates.** In this form the darshan's relationalism is recognisably the same commitment made by Mach's principle, relational interpretations of quantum mechanics, and gauge theory (only relational quantities are observable) — and it has a Yoneda-like corollary the texts also assert: a unit's "kind" (*jaati*, §6) just *is* its relational profile.

Satta then maps not to any object in the category but to the **monoidal unit** — the "empty context" *I* with which every object composes without change ($A \cong I \otimes A \cong A \otimes I$). Every one of the texts' descriptions checks against this role, clause by clause: satta "doesn't prove to be a unit" (MVD p. 74, §2 — precisely: *I* is not a system); it is formless (no internal structure); it is omnipresent and in-between (every composition of units factors through it — SB p. 79's "the definite distance between particles *is* formless existence," §4); and "saturation" is the *unitor* — an isomorphism that is always already in place, structural rather than dynamical. On this reading the magnetic-force knot dissolves: the texts reach for force vocabulary because ordinary language has no word for a structural enabling condition, but the bond's logical role is a unit law, not a dynamical arrow. "Being soaked itself is forcefulness" (SB p. 57) becomes exactly right — the bond does nothing over and above being the condition of composition.

Three further pieces fall into place under the same reading:

- **Indestructibility (§11)** is the absence of a zero object: there are no morphisms from or to nothing. Composition and decomposition are rearrangements within the tensor structure; creation *ex nihilo* and annihilation are not rare or difficult — they are ungrammatical. This is why the texts treat "nothing can be annihilated" as a logical point about counting ("every fragment is still called 'one'"), not an empirical discovery.
- **Time (§13)** is defined on morphisms — durations of activity — not as a background parameter. This is the process-first ordering that categorical approaches to physics also

adopt: processes are primary, states derivative.

- **The Five Forces table (§5) and the universal constitution principle (§6, MVD p. 149)** are claims that a structure-preserving map — a functor — exists between the insentient domain and the sentient one: same nucleus-and-orbits architecture, mapped across orders, not shared substance. This settles §15's second bullet toward "analogical," but with a precise sense of analogy that is stronger than loose metaphor: the claim is that the *structure* transfers exactly, whatever the entities are.

14.3 What the darshan asserts that physics does not: the mediative as primitive, and the development arrow. The relational axiom of 14.1–14.2 is congruent with live physics programs; it contributes an ontology, not new machinery. The mediative doctrine (§5, §8, §12) is different in kind, and it is the corpus's one structural claim that ordinary physics *denies* rather than already believes. Physics treats equilibrium as emergent (a statistical fact about many interactions) and treats the reference point of any ± opposition as conventional (a choice of zero, a gauge choice). The darshan asserts the reference is ontologically primitive: a real third alongside every generative/degenerative pair, doing continuous regulatory work ("the mediative power therefore works continuously to normalise the negative and positive charges," SB p. 70), and unmeasurable *by definition*, because measurement only registers variation and the mediative is the invariant of the variation (SB p. 250, §12). Stated physics-side, the general principle on offer is: **every observable two-valued variation presupposes a non-observable invariant that actively parameterises it.** The nearest analogues in physics are the connection/curvature distinction (curvature is measured; the connection itself is not) and renormalisation-group fixed points — but in both cases physics *derives* the invariant from the dynamics, whereas the darshan *posits* it as the more fundamental term and derives the dynamics (charge, excitation, decline, resolution) from distance to it.

KD both sharpens and strains this claim. It sharpens the *diagnosis*: physics misses the mediative because "all these studies are based on excited motion alone" (KD p. 102, §7.1) — measurement registers variation, the mediative is what holds in the unvaried state. But KD p. 103 (§8) also gives the mediative an explicitly *dynamical* job description: a two-way restoring influence that pulls orbital particles back to a set-point "good distance" against displacement in either direction. That is, formally, a restoring force about an equilibrium separation — the kind of statement physics can model (a potential well) and does measure the consequences of (bound states, vibrational spectra). So KD makes the mediative doctrine less insulated than SB leaves it: a regulator that fixes an equilibrium distance has observable signatures even if the regulator itself is "not measurable." The paper should note this as the corpus's nearest approach to a falsifiable clause — and that physics would answer: the restoring behaviour around equilibrium separations is real and well-measured, and derives from the ± interactions themselves, needing no third primitive.

The full KD translation surfaces a *second* physics-denying claim, distinct from the mediative: **the direction of spontaneous change.** Physics' second law says isolated systems drift toward equilibrium in the sense of disorder; the darshan asserts a universal drift with the opposite sign — "every excitation is understood to be working toward… natural-state motion; natural-state motion toward development progression; development… toward awakening" (KD p. 77, §8). The same monotone appears as SB p. 53's rising vibrational motion *with rising stability* (§7), §6's development progression through species toward constitutional completeness, and §9.2's Sun cooling *toward* its natural state — that last being simply this arrow applied to a star. So the corpus has two teleological primitives, and they compose: the mediative supplies the local restoring tendency (equilibration about set-points), and the development arrow supplies the global drift (successive set-points ascending toward completeness). A physics reader should be told plainly that this arrow runs against the entropic one, and that the darshan would answer the obvious objection (order increases locally only by exporting disorder) by denying the bookkeeping premise — for it, de-excitation is not export of disorder but approach to the natural state, which costs nothing.

Two liabilities, flagged for the paper: (i) the unmeasurability doctrine insulates the mediative claim from every empirical test — a physics reader will identify this immediately, and the paper should concede it rather than soften it (with the KD qualification above); (ii) §11's conservation of *units* (countability) conflicts with particle creation and annihilation in quantum field theory unless the darshan's "particles" sit below field quanta — a question the texts never face (no antimatter, no field-quantum vocabulary anywhere in the corpus).

14.4 Reference map. A rough map, for reference:

Physics vocabulary	Primary-text location	Level assigned
Pressure, attraction, repulsion	SB p. 49	Unit mutuality
Heat, sound, electricity, waves, fields ("relative energy")	MVD p. 40	Unit mutuality
Magnetism / electromagnetic force	SB pp. 85, 128; KD pp. 54, 103, 118	Unit mutuality (named force); also used for the saturation bond itself (§4)
Electricity ("fragmentation of magnetic current"), waves, sound	KD pp. 118–119	Unit mutuality (medium-borne; all nature electricity-receptive)
Gravitational force / weight	JV pp. 149–150; MVD p. 45; KD pp. 103–104	Unit mutuality (tendency to combine; "evidencing coexistence with Earth" — KD bounds it to Earth's environment)
Weak / strong interaction	SB p. 85	Unit mutuality (atomic-scale, table only — no further gloss found)

Physics vocabulary	Primary-text location	Level assigned
"Mediative force"	SB pp. 84–85; KD pp. 102–103	Unique to darshan; atomic-nucleus-level, regulates the other four (KD: two-way restoring toward "good distance")
Atomic structure (nucleus + orbits, "kind"/element)	MVD pp. 42, 78, 149; AVD pp. 103, 185; KD pp. 53–57	Unit-internal (particle count and arrangement; KD: nucleus ≥ orbits, four-orbit equilibrium, 108 species acknowledged)
Charge / excitation (*āvesh* — context-dependent, §8)	SB pp. 70, 79, 88; AVD p. 103; KD pp. 57, 74, 118	Unit-internal: ± charge \= generative/degenerative imbalance (orbit-to-nucleus distance); excited state \= the disturbed condition it produces (KD: precondition of particle transfer)
Mass (atomic)	SB pp. 86, 182; AVD p. 185; KD p. 56	Unit-internal (particle count in nucleus + orbits; KD keys weight to nucleus count)
States of matter (solid, liquid, gaseous)	SB pp. 97, 142, 145; MVD pp. 41–42; JV p. 14	Unit composition (three states only; no plasma)
Radioactivity / nuclear explosion	SB pp. 258–259	Unit-internal (overfull atoms; mediative regulation overwhelmed)
Density (with shape, volume: form)	SB pp. 65, 86, 119, 249–250; MVD p. 50; AVD pp. 185, 199	Unit-internal (the boundary/framework of a unit)
Indestructibility (no creation, no annihilation)	SB pp. 49–50, 103; AVD pp. 69–70, 245, 247–248; JVD p. 82; JV p. 125; MVD p. 299	Dharma of every unit (conservation analogue — units, not quantities)
Cosmogony (Big Bang, rejected)	JV p. 149	Denied of satta (forceless ground); no origin event needed
Celestial mechanics (elliptical orbits, seasons, solar heat)	KD p. 76; SB pp. 252, 259	Unit mutuality ("good distance"; teleological — order for the sake of seasons/development)
Time	MVD p. 34; SB p. 48; KD p. 134	Unit activity (KD: clock-time a derived convention; "time became primary, the object forgotten")

Physics vocabulary	Primary-text location	Level assigned
Development arrow (excitation → natural-state → development → awakening)	KD p. 77; SB pp. 53, 62; MVD p. 149	Universal directional drift of unit-states — opposite in sign to the thermodynamic arrow (§14.3)
Absolute / uniform energy	MVD pp. 40, 45; SB pp. 48–50	Satta itself — not a force, the ground of all the above
Forcefulness / "magnetic forcefulness" of saturation	SB pp. 57, 61–62, 79, 98	The satta–unit bond itself (endowment, not efficient causation)

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15. Open interpretive tensions (flag for the paper, not yet resolved)

- **"Magnetic force" vs. actionless doctrine.** SB literally calls the saturation bond's forcefulness "magnetic force" (चुम्बकीय बल, p. 61) while also insisting *satta* is actionless and non-transforming. The texts do not reconcile these in the same passage. The most charitable reading (used in *The Ontology of Coexistence* §1.2) is that "magnetic force" names a standing, eternal endowment rather than a transmitted quantity — but this is an interpretive choice, not something the text states outright. §14.2's formal reading (saturation as unit law, not dynamical arrow) makes the choice principled rather than merely charitable. Worth a dedicated footnote or subsection wherever this note's material gets folded into a published study.
- **Five Forces table — descriptive or analogical?** SB p. 85 does not say whether it is claiming physics is incomplete without a fifth ("mediative") force, or simply using the four-force inventory as a structural analogy for jeevan's five faculties. The strong/weak interaction rows get no independent gloss anywhere else in the corpus (only the table itself) — thin evidentiary support compared to electromagnetic/gravitational, which are discussed extensively elsewhere. §14.2's functor reading favours the analogical horn, with a precise sense of "analogy" (structure transfers exactly; substance does not).
- ***Āvesh* as evaluative vocabulary.** SB ties charge/excitation definitionally close to "problem" and its absence to "resolution" — useful for the book's thesis (Resolution-Centred Materialism) but worth flagging as a place where physics vocabulary is doing double duty as ethical/soteriological vocabulary; MD-Mapping's own definition of *āvesh* ("deluded conduct-tendency; motion toward decline") builds the evaluation into the word. The contextual rule (§8: ± imbalance \= charge; disturbed condition \= excited state) resolves most instances, but a reader could still ask whether *āvesh* is one concept

with two faces or two concepts sharing one word — the texts treat the ± imbalance and the disturbed state as one phenomenon, which is itself a substantive claim.

- **No subatomic-particle vocabulary, no periodic table, no atomic numbers.** Electron, proton, neutron do not appear in SB, MVD, JV, AVD, JVD, or KD. "Particles" (*ansh*) are counted generically, and no named element (hydrogen, iron, gold, etc.) is ever given a specific particle count — only a growth rule (§6) now supplemented by KD's structural constraints (§6.4a: minimum 1+1, nucleus ≥ orbits, four-orbit equilibrium) and KD p. 55's acknowledgement of the "108 species" inventory as a human claim. Any comparison to modern atomic physics needs to keep the remaining gap explicit: the darshan has the *shape* of a shell model, and now some of its *inequalities*, but still no per-element numbers (no analogue of "carbon \= 6"). The one place a specific, named, fixed structure does appear is *jeevan* itself — exactly four orbits plus nucleus, ten activities, psychologically named rather than numbered (§6.4) — which is a special case, not the general insentient-atom model. (Curiously, KD's nucleus ≥ orbits inequality *holds* under the natural modern mapping: if orbital *ansh* \= electrons and *madhyansh* \= nucleons, then "at least as many in the middle as in the orbits, possibly more" is exactly $A \geq Z$ — true of every neutral atom, with equality only at hydrogen-1, which is also KD's stated minimum atom (1+1). Whether this is insight or coincidence cannot be settled from the texts, since *ansh* is never identified with any named particle — but it is the corpus's single most striking numerical agreement with standard atomic physics and deserves a paragraph in the paper.)
- **"Overfull" is undefined.** SB p. 259 rests radioactivity (and the entire solar model of §9.2) on atoms being "overfull," but no text says overfull relative to *what* — there is no stated maximum particle count per orbit or per nucleus (consistent with §6's missing numbers, but here the gap is doing explanatory work). Also unreconciled: §9.2's Sun-as-cooling-toward-natural-state reverses the direction of standard stellar physics, and the texts never engage fusion as an energy source at all.
- **Jeevan's four orbits vs. insentient atoms' open-ended orbit count — now partially answered by KD.** MVD p. 78 allows ordinary atoms "first, second, third, fourth, and so on" orbits (unspecified upper bound) but fixes *jeevan* at exactly four, without explanation. KD p. 56 (§6.4a) supplies what looks like the missing law: orbit counts above four tend to decrease and below four tend to increase — four is the equilibrium of the orbit-count dynamics. On that reading, constitutional completeness is the *stable fixed point* of atomic development, and jeevan's four orbits are not an arbitrary cap but the attractor every developing atom moves toward — which meshes exactly with §6.4b's *prasthāpan–visthāpan* (the particle-transfer process that moves atoms through species) and with MVD p. 78's open-ended counts for atoms still en route. Still open: *why* four is the equilibrium (KD asserts the tendency, not its ground), and whether the §1.6.1 (*gathanpurnata*) material glosses completeness as specifically the four-orbit configuration.

- **KD’s gravitation is local, not universal.** KD p. 104 bounds the falling-tendency “up to this Earth’s environment,” with each body “complete with its own environment” (§10.5). Taken at face value this denies universal gravitation — yet §9.4 (KD p. 76) has the planets maintaining “good distance” around the Sun, which universal gravitation explains and a strictly Earth-bounded tendency does not. The texts do not connect the two passages. Options for the paper: (a) “good distance” order among planets is a separate, non-gravitational principle for KD (consistent with JV’s zero-attraction doctrine, §10.1); (b) the boundedness claim concerns only the *falling* arrangement, not inter-body order. The corpus underdetermines the choice.

Method note (for reproducibility)

Primary texts were converted to plain text with `pdftotext -layout`, and page numbers were cross-checked two ways: (1) internal footer markers (“page-NN”) embedded in the MVD and JVD PDFs, present roughly every page; (2) for SB and JV, where footers were not machine-readable, page indices were calibrated against page citations already established and used consistently in *The Ontology of Coexistence* (e.g. “SB, p. 48,” “SB, p. 62,” “MVD, p. 40”) by locating the matching English sentence and checking the offset was constant (confirmed: PDF page-splitting index + 1 \= printed page number, holding across more than five independent check-points in SB). Quotes above are transcribed from the English portion of the bilingual PDFs; Hindi originals were consulted for the technical terms glossed in brackets but are not reproduced here except where already given in the study’s own convention (italic transliteration).

KD (Manav Karm Darshan, v5) requires a separate protocol. The PDF is Hindi-only (no English translation exists in the References), and its embedded font’s character map is corrupted, so machine-extracted text is systematically garbled (e.g. त/स्/ि cycle, श↔र्) — the text layer was used only to *locate* candidate pages by cipher-adjusted term search, never to transcribe. All KD passages cited here were read visually from rendered page images and translated by us. **Full working translations of chapters 3.1–3.3, 3.7–3.8, and 3.11–3.15 now exist at `References/Madhyasth-Darshan/KD-English/`** (with `[p. NN]` printed-page markers, a README index, and a glossary of term decisions in `KD-Glossary-Additions.md` + `KD-Translation-Glossary.xlsx`); inline KD quotes in this note should be read against those files, which supersede them where wording differs, and all remain **working translations pending a published English edition**. Page numbers cited are the printed page numbers in the running headers (calibration: printed page \= PDF page − 25, constant across all pages checked). Two fixed terminology conventions: *āvesh/āveshit* is translated **contextually** — “charge” where the text concerns the positive/negative (generative/degenerative) imbalance of atoms, “excited/excitation” where it concerns a unit’s disturbed state as against its neutral/natural state (rule stated in full in §8’s terminology note; SB’s published English uses “charge” for both and glosses the state sense “excited (charged)” at p. 257; MD-Mapping carries both readings); and *śram* \= **effort** (MD-

Mapping's notes: "We are using 'effort' for shram in the sense of shram-gati-parinam"), with "labour" reserved for economic compounds (श्रम मूल्य \= labour value). Coverage status: the physics-relevant chapters of KD section 3 (chapters 3.1–3.15, printed pp. 53–135) are now fully reviewed except **3.9 (quantity, *mātrā*) and 3.10 (properties, *guṇa*)**, which remain untranslated and are the one known gap worth a dedicated pass (their titles overlap §12's form/measurement material); chapters 3.4–3.6 concern mental wellbeing and awakening, not physics; ch. 3.12 (reflection–projection) proved on full translation to be chiefly epistemological — teaching/study as reflection–projection — with heat transfer as one species of the general category (noted in §9.4).